

Q1: What is Software Quality Assurance (SQA)?

Software Quality Assurance (SQA) is a systematic process that ensures software is developed according to established quality standards, organizational processes, and customer requirements. It focuses on preventing defects by improving the software development process rather than only finding bugs after the software is built.

Why is Quality Important in Software Development?

Software quality is essential because it ensures that the final product is reliable, secure, efficient, and meets customer expectations. The importance of software quality includes:

1. **Prevent Defects** – Identify and prevent defects early through reviews, coding standards, and static analysis.
2. **Improve Customer Satisfaction** – Deliver reliable, user-friendly software that meets customer needs and expectations.
3. **Reduce Development Cost** – Detect defects early to minimize rework, maintenance, and expensive post-release fixes.
4. **Risk Reduction** – Reduce business, security, and operational risks before software is released.
5. **Process Improvement** – Continuously improve development and testing processes using defect analysis and quality metrics.
6. **Compliance** – Ensure the software complies with legal, regulatory, and industry standards, especially in sectors like banking, healthcare, and fintech.

Q2: Differentiate between QA VS QC VS Testing.

Aspect	Quality Assurance(QA)	Quality Control (QC)	Testing
Focus	Focuses on improving the development process to prevent defects.	Focuses on evaluating the software product to identify defects.	Focuses on executing test cases to verify that the software works as expected.
Goal	Prevent defects before they occur by improving processes and standards.	Detect defects in the developed product before release.	Find bugs and verify that the software meets the specified requirements.
Approach	Proactive – Prevents defects by establishing quality processes.	Reactive – Detects defects after the product is developed.	Reactive – Detects defects after the product is developed.

Activities	Process audits, quality planning, defining standards, process reviews, training, and continuous process improvement.	Product inspections, code reviews, document reviews, and verifying whether the product meets quality standards.	Writing and executing test cases, exploratory testing, regression testing, automation testing, and reporting defects.
Responsibility	Shared responsibility of the entire development team, including developers, testers, managers, and stakeholders.	Mainly performed by the QA/QC team responsible for product quality verification.	Performed by Test Engineers, QA Engineers, or Automation Engineers.
Example	Creating coding standards and defining a mandatory code review process before development.	Reviewing the completed software build to ensure it complies with the defined standards and quality checklist.	Executing login, registration, or checkout test cases on the latest software build and reporting any defects found.

Q3: How is AI changing Software Testing?

Artificial Intelligence (AI) is transforming software testing by making it more intelligent, adaptive, and efficient. Unlike traditional testing, AI can learn from historical data, adapt to application changes, and assist throughout the entire Software Development Life Cycle (SDLC).

1. From scripted to Intelligent Testing

Traditional automation relies on fixed test scripts that must be manually updated whenever the application changes. AI-driven testing understands the intent behind test cases, adapts to changes automatically, and learns from previous test executions, making automation more flexible and reliable.

2. AI across the Software Development Life Cycle (SDLC)

AI is no longer limited to test execution. It supports multiple stages of the SDLC, including:

- Requirement analysis
- Test case generation
- Code review
- Test execution
- Defect analysis and triage

3. Self-Healing Test Automation

One of the most valuable AI features is self-healing automation. When UI elements or locators change, AI automatically identifies the updated element and repairs the test script without requiring manual intervention. This significantly reduces test maintenance and improves automation stability.

4. Intelligent Reporting and Defect Analysis

Instead of providing only pass/fail results, AI generates intelligent reports by:

- Summarizing test failures
- Grouping similar defects
- Identifying patterns in failures
- Highlighting possible root causes

This enables QA teams to analyze issues more quickly and make better decisions.

Q4: Explain the responsibilities of QA Engineer, Software Test Engineer, AI Quality Engineer with daily activities of each role.

1. QA Engineer

Ensures that the software development process follows quality standards and best practices. Focuses on preventing defects and improving development processes.

Daily Activities

- Participate in requirement review meetings.
- Create test plans and test strategies.
- Review requirements and test cases.
- Monitor QA processes and quality standards.
- Collaborate with developers and stakeholders.
- Track defects and ensure process improvements.

2. Software Test Engineer

Verifies that the software functions correctly by identifying defects through manual and automated testing. Ensures the product meets functional and non-functional requirements.

Daily Activities

- Design and execute test cases.
- Perform manual and automation testing.
- Log and track defects in bug-tracking tools.
- Perform regression, smoke, and integration testing.
- Verify bug fixes and retest applications.
- Prepare test execution reports.

3. AI Quality Engineer

Ensures the quality, reliability, fairness, and safety of AI/ML systems. Validates AI models, monitors their performance, and checks for bias and ethical issues.

Daily Activities

- Test AI/ML models using different datasets.
- Evaluate model accuracy, precision, and recall.
- Detect bias, fairness, and ethical issues.
- Monitor AI model performance after deployment.
- Validate AI-generated outputs.
- Work with data scientists and ML engineers to improve model quality.

Q5: Draw the career roadmap of a QA Professional.

